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GRADE A+
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Master of Computer Applications
Semester – III
Data Analytics for Business
Experiment No.: 07
Title: Developing finance KPI dashboards

Submitted By:

Name: GANESH A

Reg. No.: 2511022250015

Date: 15-08-2026

Submitted To:

Faculty Name: Mr. Aman Kumar Sharma

Department of Computer Application
Alliance University
Chandapura - Anekal Main Road, Anekal
Bengaluru – 562106

Experiment No.: 07

Developing finance KPI dashboards

Problem Statement

You are a Financial Analyst in a manufacturing company. Senior management wants to track revenue, expenses, profit margins, and budget utilization. Create a finance KPI dashboard that helps evaluate the financial performance of the organization.

Methodology / Steps Followed

Step 1: Data Collection

- Obtained Finance Analysis data from Kaggle datasets.

Step 2: Data Import

- Opened Power BI Desktop.
- Imported all datasets using the **Get Data** option.
- Loaded CSV file into Power BI.

Step 3: Data Cleaning and Transformation

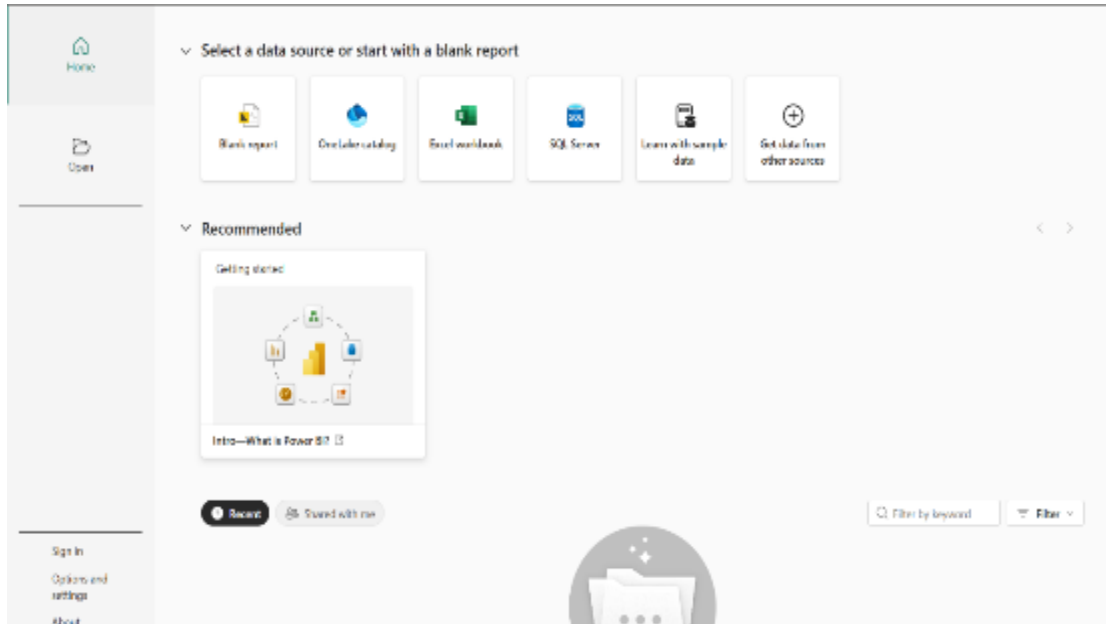
- Removed duplicate records.
- Handled missing values.
- Converted data types where necessary

Step 4: Dashboard Development

- Created KPI Cards to display Total Budget, Total Actual Expense, Budget Variance, Budget Utilization, and Over-Budget Transactions.
- Developed a Clustered Column Chart to compare Budget and Actual Expenses across different Departments.
- Developed a Line Chart to visualize monthly Budget vs. Actual Expense trends over time.
- Created a Bar Chart to analyse Actual Expenses across different Categories.
- Created a Donut Chart to analyse the overall distribution of Expenses by Category.
- Developed a Clustered Column Chart to compare Budget and Actual Expenses across different Regions.
- Created a Bar Chart to analyze Actual Expenses based on different Payment Methods.
- Developed a Budget Status Donut Chart to compare the proportion of Within-Budget and Over-Budget transactions.
- Added Interactive Slicers for Date, Department, Region, Category, and Payment Method to enable dynamic financial analysis.

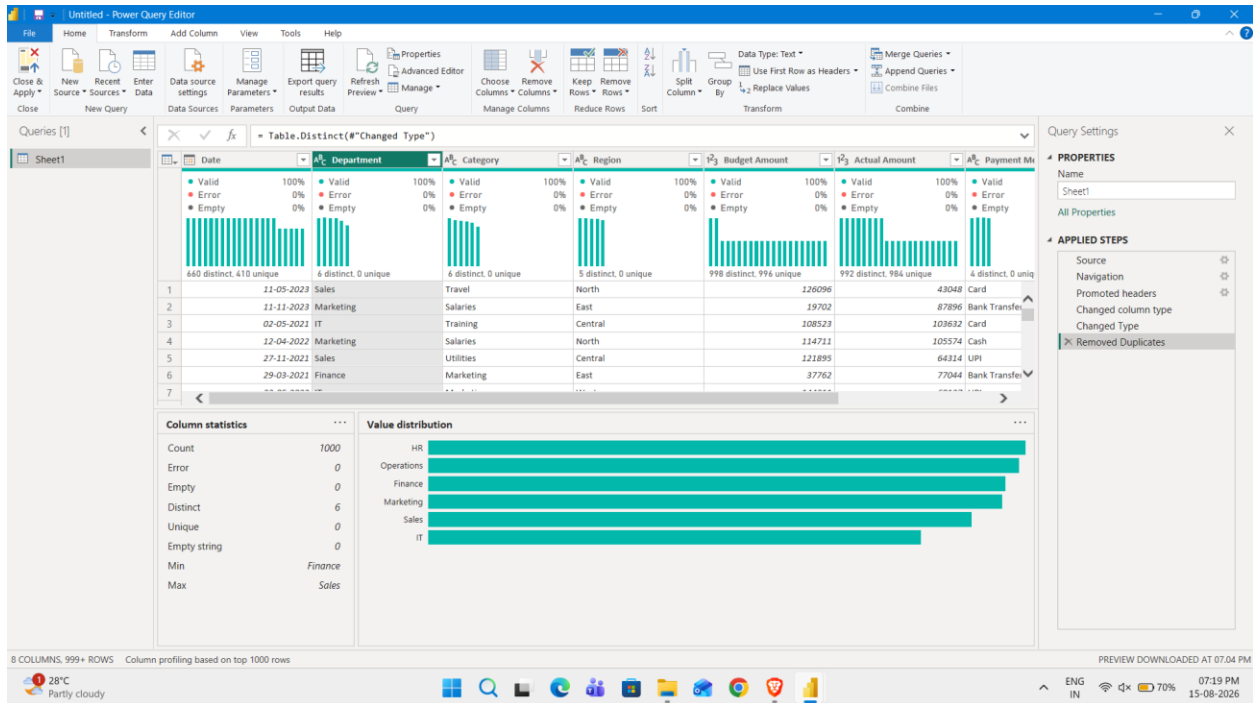
Output

- Opened Power BI file.

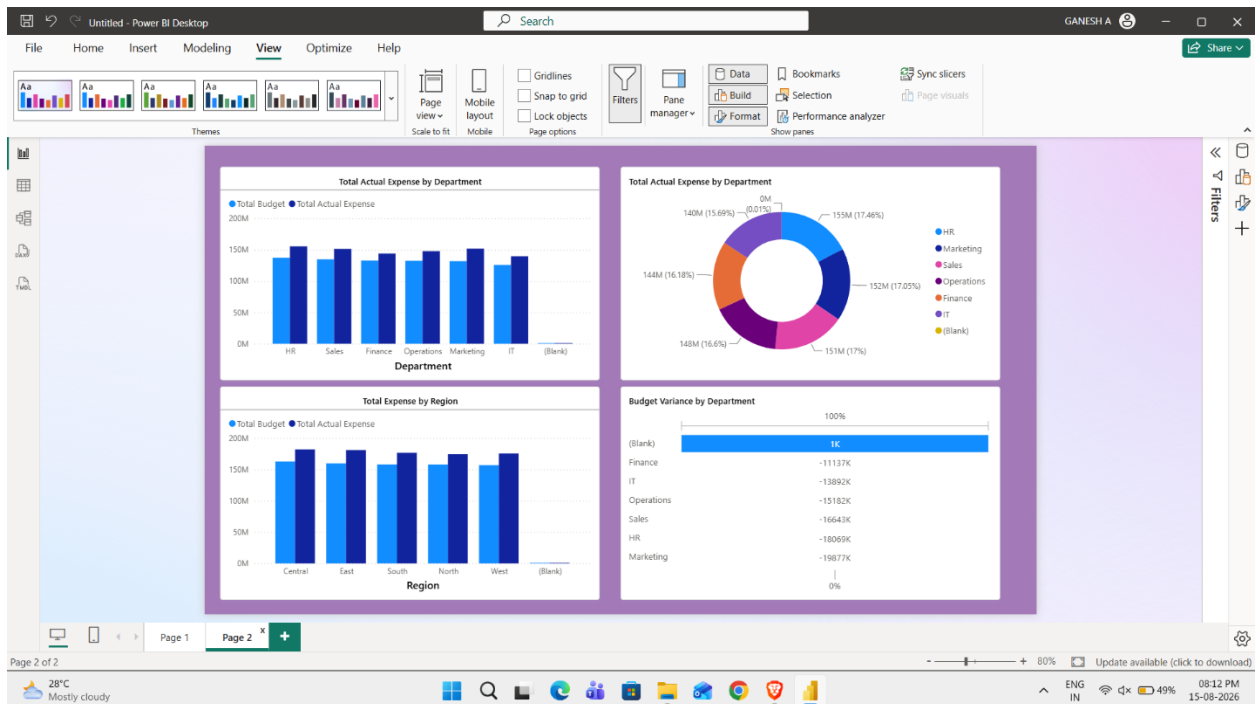
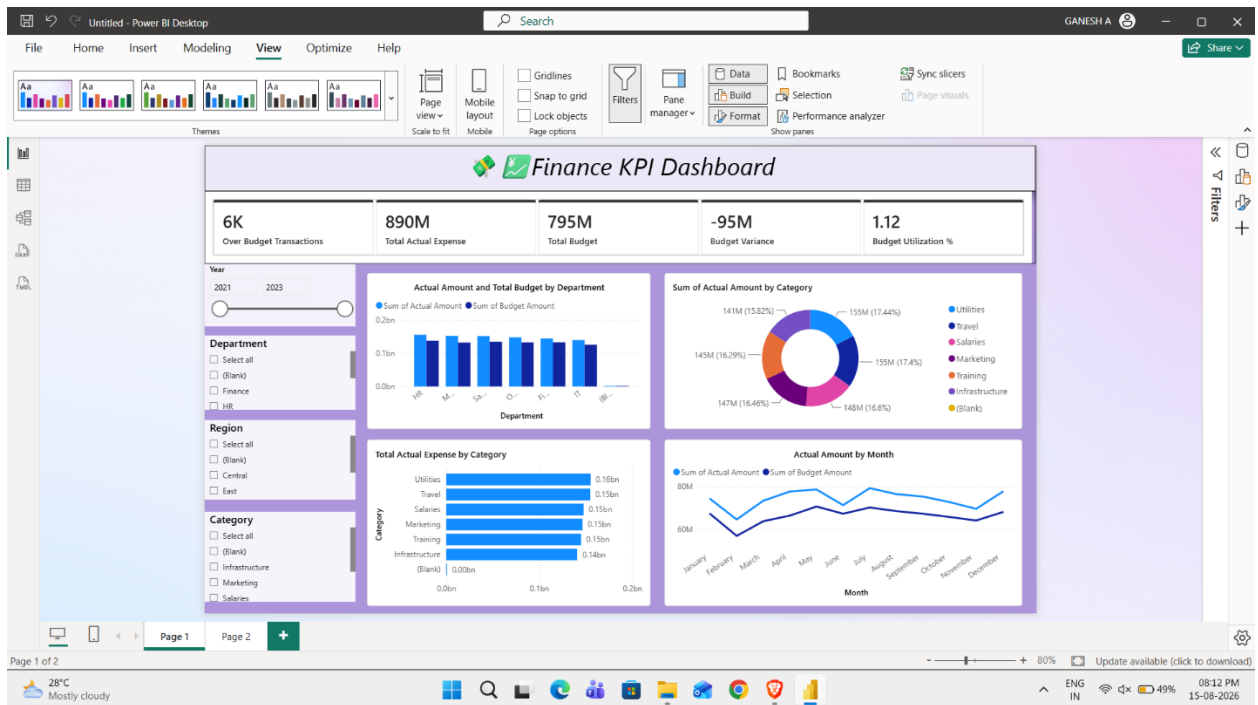


- Loaded the dataset.
- Transformed data And applied Cleaning where ever Necessary.

Date	Department	Category	Region	Budget Amount	Actual Amount	Payment Method
11-05-2023	Sales	Travel	North	126096	43048	Card
11-12-2023	Marketing	Salaries	East	19702	87896	Bank Transfer
02-05-2021	IT	Training	Central	108523	103632	Card
12-04-2022	Marketing	Salaries	North	114711	105374	Cash
27-11-2021	Sales	Utilities	Central	121895	64314	UPI
29-03-2021	Finance	Marketing	East	37762	77044	Bank Transfer
22-05-2023	IT	Marketing	West	144321	69127	UPI
11-05-2021	HR	Marketing	Central	22567	53015	Cash
09-02-2023	IT	Infrastructure	West	95737	58794	UPI
10-12-2021	IT	Salaries	East	16857	253432	Cash
17-03-2023	IT	Utilities	East	79603	73056	Bank Transfer
21-01-2022	Finance	Utilities	Central	85399	109122	Cash
14-08-2023	Sales	Marketing	Central	68086	70502	Bank Transfer
04-10-2021	HR	Training	East	99145	51102	UPI
05-04-2022	Sales	Infrastructure	Central	42270	148677	Cash
22-01-2021	Marketing	Infrastructure	Central	130071	79899	Card
10-09-2021	Marketing	Marketing	Central	12012	99017	Card
18-01-2023	Marketing	Marketing	Central	69049	120109	Card
07-05-2023	Sales	Salaries	North	94989	30233	Bank Transfer
20-04-2022	Operations	Training	East	99586	30720	Cash
19-12-2023	Finance	Infrastructure	East	49080	167039	Card



- Added Dax Queries to the KPI Cards
- Average Expense =
AVERAGE('Sheet 1'[Actual Amount])
- Within Budget Transactions =
CALCULATE(COUNTROWS('Sheet 1'),'Sheet 1'[Actual Amount] <='Sheet 1'[Budget Amount])
- Budget Variance = [Total Budget] - [Total Actual Expense]



Result

- Successfully developed an interactive Finance KPI Dashboard using Power BI to support financial performance monitoring and budget management.
- Enabled management to monitor key financial metrics including Total Budget, Actual Expenses, Budget Variance, Budget Utilization, and Over-Budget Transactions.
- Provided clear insights into department-wise, category-wise, and region-wise spending patterns to identify areas of high expenditure.
- Analyzed monthly budget and actual expense trends to identify periods of increased spending and potential budget overruns.
- Evaluated budget performance across regions and departments to support better financial control and resource allocation.
- Implemented interactive slicers for Date, Department, Region, Category, and Payment Method to enable dynamic filtering and cross-analysis.
- Created a user-friendly two-page executive dashboard that supports data-driven financial decision-making, budget control, expense monitoring, and management planning.
- Used Power Query for data cleaning and transformation and DAX measures for KPI calculations, ensuring accurate and interactive financial analysis.

Conclusion

The Finance KPI Dashboard was successfully developed using Power BI to provide a clear and interactive view of the organization's financial performance. Power Query was used to clean and transform the financial dataset, while DAX measures were implemented to calculate important KPIs such as total budget, actual expenses, budget variance, and budget utilization. The two-page dashboard provides management with useful insights into department-wise, category-wise, and region-wise expenses, as well as monthly spending trends and budget performance. Interactive slicers make it easier to explore the data and identify areas of overspending.

Overall, the dashboard provides a user-friendly and data-driven solution for financial monitoring, budget control, expense analysis, and management decision-making. It can also be further enhanced by adding revenue and profit-related data in the future to provide a more complete view of financial performance.

Learning Outcomes

Through this project, I learned:

- Gained practical knowledge of **Power BI** for developing interactive and professional financial dashboards.
- Learned how to use **Power Query** for data cleaning, transformation, handling missing values, and removing duplicate records.
- Developed an understanding of **DAX** for creating calculated measures and financial KPIs.

- Learned to calculate and analyze **Total Budget, Actual Expense, Budget Variance, and Budget Utilization**.
- Gained experience in creating different **data visualizations**, including KPI cards, bar charts, column charts, line charts, donut charts, and waterfall/scatter charts.
- Learned how to analyze financial performance across **departments, categories, regions, and payment methods**.
- Gained experience in using **interactive slicers and filters** for dynamic data analysis.
- Improved understanding of identifying **budget overruns and spending trends** through visual analytics.
- Developed skills in presenting complex financial data in a **simple and management-friendly format**.
- Gained practical experience in using **data-driven insights to support financial planning and decision-making**.

Github Link:

https://github.com/GANESHA08/Data_Analytics_projects/tree/main/Week%207%20Experiment